

Question

Write a Java program to demonstrate single inheritance using a parent class and a child class.

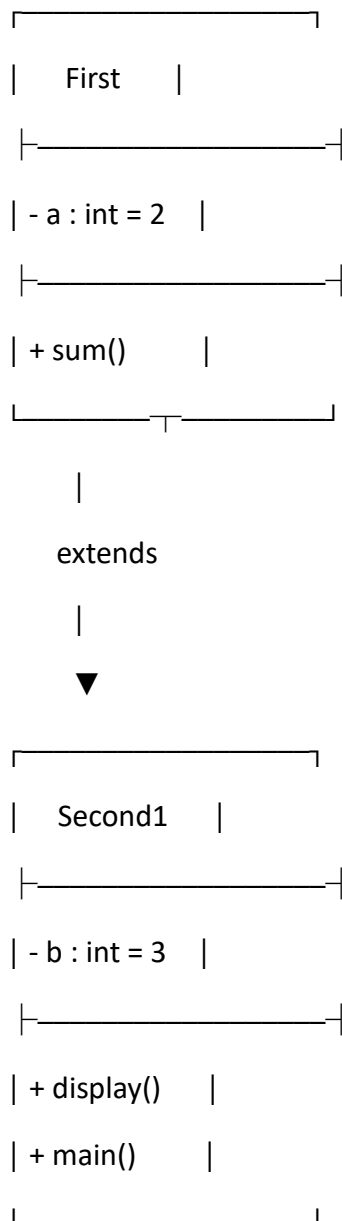
Steps for Execution

1. Create a parent class First.
2. Declare an integer variable $a = 2$.
3. Define the `sum()` method to calculate $a + 5$.
4. Create a child class Second1 using `extends First`.
5. Declare an integer variable $b = 3$.
6. Define the `display()` method to calculate $b + 8$.
7. Create an object `s` of class Second1.
8. Call `display()` using the object.
9. Call the inherited `sum()` method.
10. Access the inherited variable `a`.
11. Display all results.

Algorithm

12. **Start**
13. Define class First.
14. Initialize $a = 2$.
15. Define `sum()` and print $a + 5$.
16. Define class Second1 extending First.
17. Initialize $b = 3$.
18. Define `display()` and print $b + 8$.
19. Create an object of Second1.
20. Call `display()`.
21. Call inherited method `sum()`.
22. Display inherited variable `a`.
23. **Stop**

UML Class Diagram



Code

Writing

```
package javaprograms;
```

```
class First {
```

```
    int a = 2;
```

```
    void sum() {
```

```
        System.out.println("sum is:" + (a + 5));
    }
}
```

```
class Second1 extends First {
    int b = 3;

    void display() {
        System.out.println("sum is:" + (b + 8));
    }

    public static void main(String[] args) {
        Second1 s = new Second1();

        s.display();
        s.sum();
        System.out.println("variable is:" + s.a);
    }
}
```

Output

sum is:11

sum is:7

variable is:2